Application Overview – Music Playlist Web App

Intended Users:

- - General music listeners
- - System administrators managing the music database

Problem Addressed:

- - Users need a simple way to organise favourite songs into personalised playlists
- - Music data needs centralised and structured management
- - Users lack a unified platform to discover, listen to, and discuss music.

What the Application Does:

- - Browse and search songs stored in a database
- - Create and manage personal playlists
- - Secure user authentication for personalised data
- - View browsing history and interact via comments.
- - Admin management of songs and users

Value Provided:

- - Easy and structured music organisation
- - Demonstrates authentication, database interaction, and role-based control
- - Seamless playback experience and community interaction

MoSCoW User Stories

Must Have:

As a user,

1. I want to register an account,
   so that I can have a personalised profile and save my playlists.
2. I want to log into the system,
   so that I can securely access my personal music data.
3. I want to browse the music library,
   so that I can discover available songs stored in the database.
4. I want to search for songs using keywords,
   so that I can quickly find music I am interested in.
5. I want to create a playlist by entering a name and description,
   so that I can organise my favourite songs.
6. I want to add or remove songs from my playlists,
   so that I can customise my music collections.
7. I want to stream music directly within the application (Play/Pause, Next/Prev, Progress bar), so that I can listen to songs without leaving the site,
8. I want to view my browsing history, so that I can easily recall and find songs I listened to recently.

As an administrator,

- I want to add, edit, or delete songs in the system,
  so that the music database remains accurate and up to date.

Should Have:

As a user,

- I want to filter songs by genre or artist,
  so that I can explore music more efficiently.
- I want to post comments on specific songs, so that I can share my thoughts and interact with the community.
- I want to view song details (lyrics, artist info, metadata), so that I can learn more about the music

As an administrator,

- I want to view and manage user accounts,
  so that inappropriate or inactive accounts can be controlled.

Could Have:

As a user,

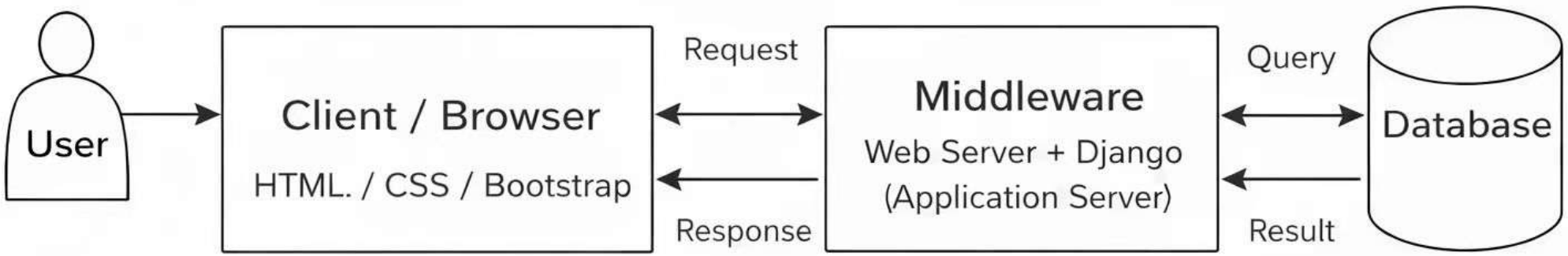
- I want to favourite songs,
  so that I can quickly access my preferred music.
- I want to rate songs,
  so that I can express my preferences and influence recommendations.
- want to receive song recommendations based on my playlists,
  so that I can discover new music more easily.

Won’t Have:

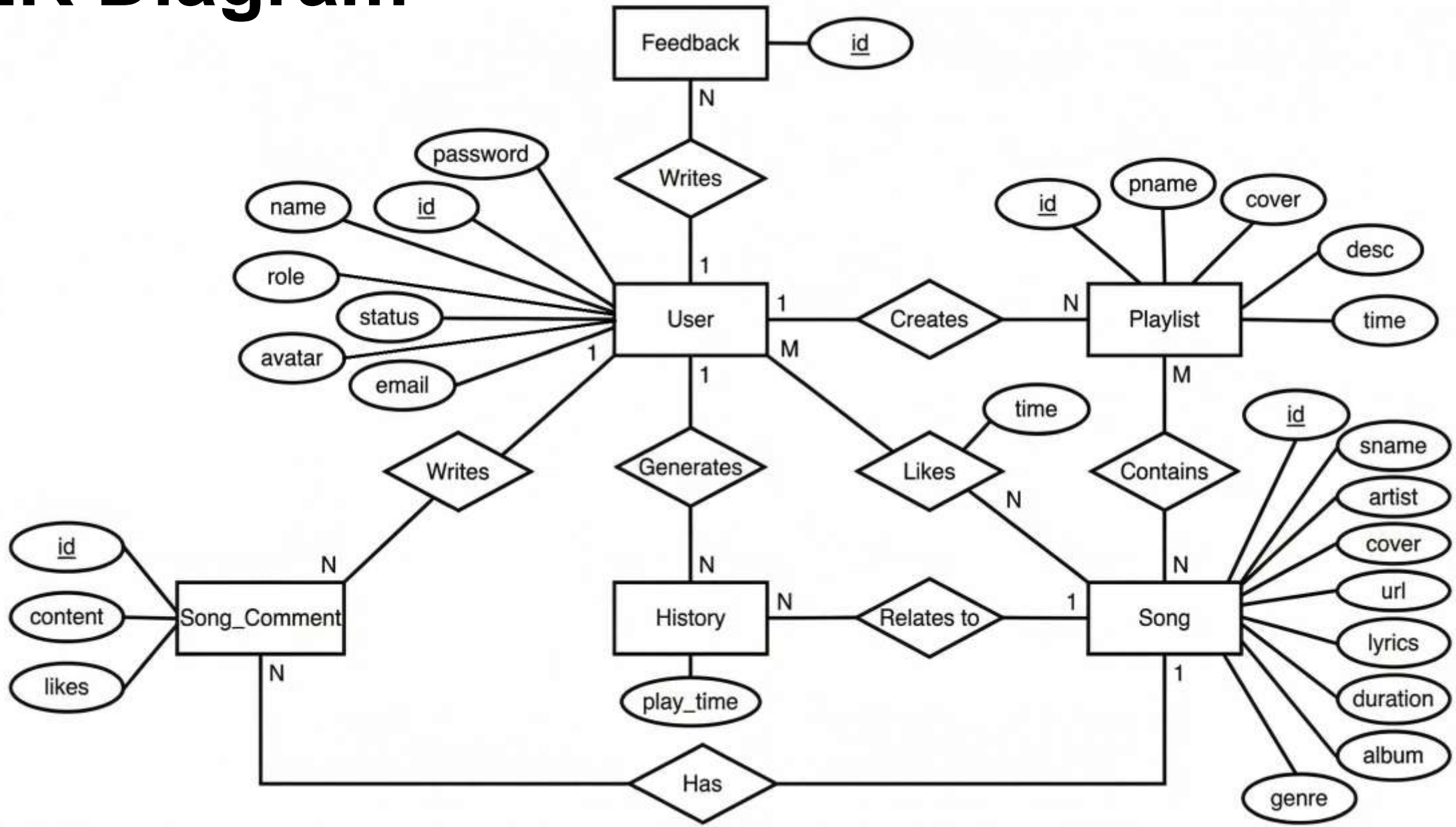
As a user,

- I want to connect my Spotify or Apple Music account,
  so that I can import my existing playlists.

Music Hub – System Architecture



ER Diagram


Feedback
id
N
Writes
1
password
name
id
role
status
avatar
email
User
1
1
M
Creates
N
Playlist
id
pname
cover
desc
time
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Contains
N
Writes
N
Generates
N
Likes
time
N
id
sname
artist
cover
url
lyrics
duration
album
genre
Song
1
1
id
content
likes
Song_Comment
N
N
History
N
Relates to
1
play_time
Has

Entity-Relationship Attribute Description

User

- **id (PK):** Unique identifier for each registered user
- **name:** User's display name shown on profile and comments
- **email:** Unique login credential used for authentication
- **password:** Encrypted password for secure access
- **role:** Defines permission level (e.g., user, admin)
- **avatar:** URL of the user's profile image
- **status:** Account state indicating whether the user is active or banned

Song

- **id (PK):** Unique identifier for each song
- **sname:** Title of the song
- **artist:** Name of the performing artist or band
- **album:** Album to which the song belongs
- **cover:** URL of the album artwork
- **url:** Location of the audio file for streaming
- **lyrics:** Text content of the song lyrics
- **genre:** Musical category of the song
- **duration:** Length of the song

Playlist

- **id (PK):** Unique identifier for each playlist
- **pname:** Name of the playlist
- **desc:** Description of the playlist content or theme
- **cover:** URL of the playlist cover image
- **time:** Timestamp indicating when the playlist was created

Song_Comment

- **id (PK):** Unique identifier for each comment
- **content:** Text of the user's comment
- **likes:** Number of likes received by the comment

History

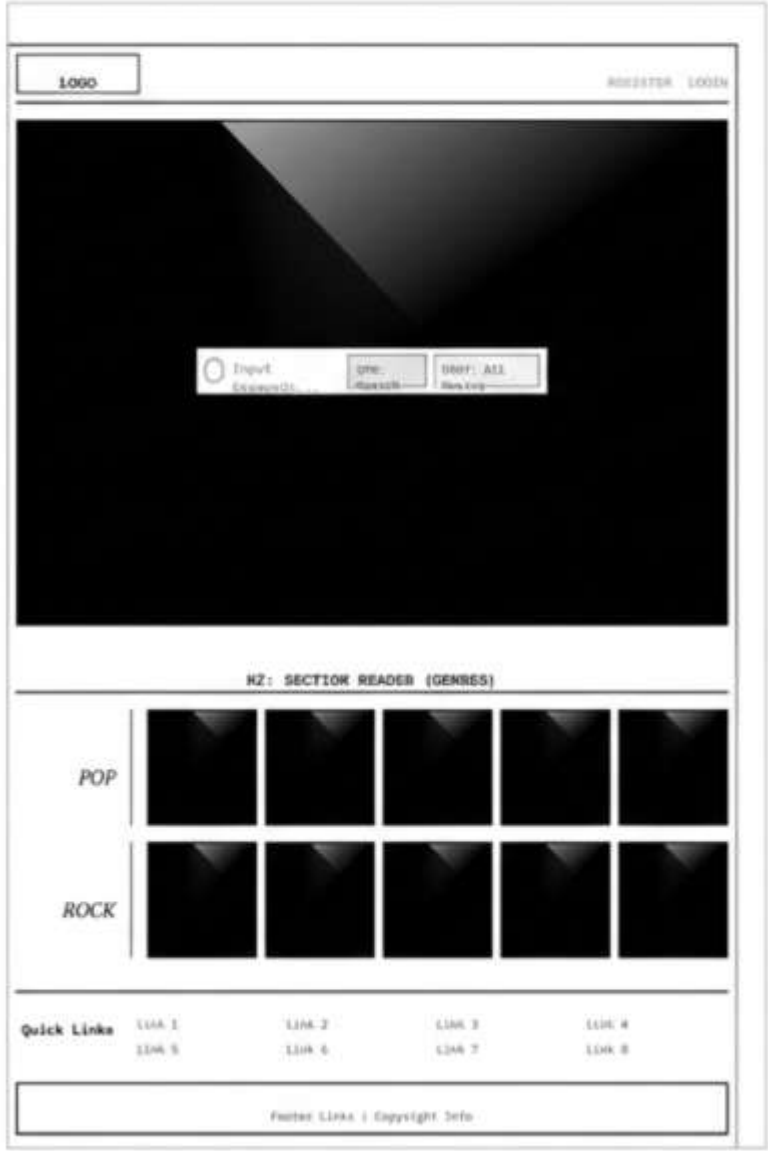
- **play_time:** Timestamp recording when a song was played

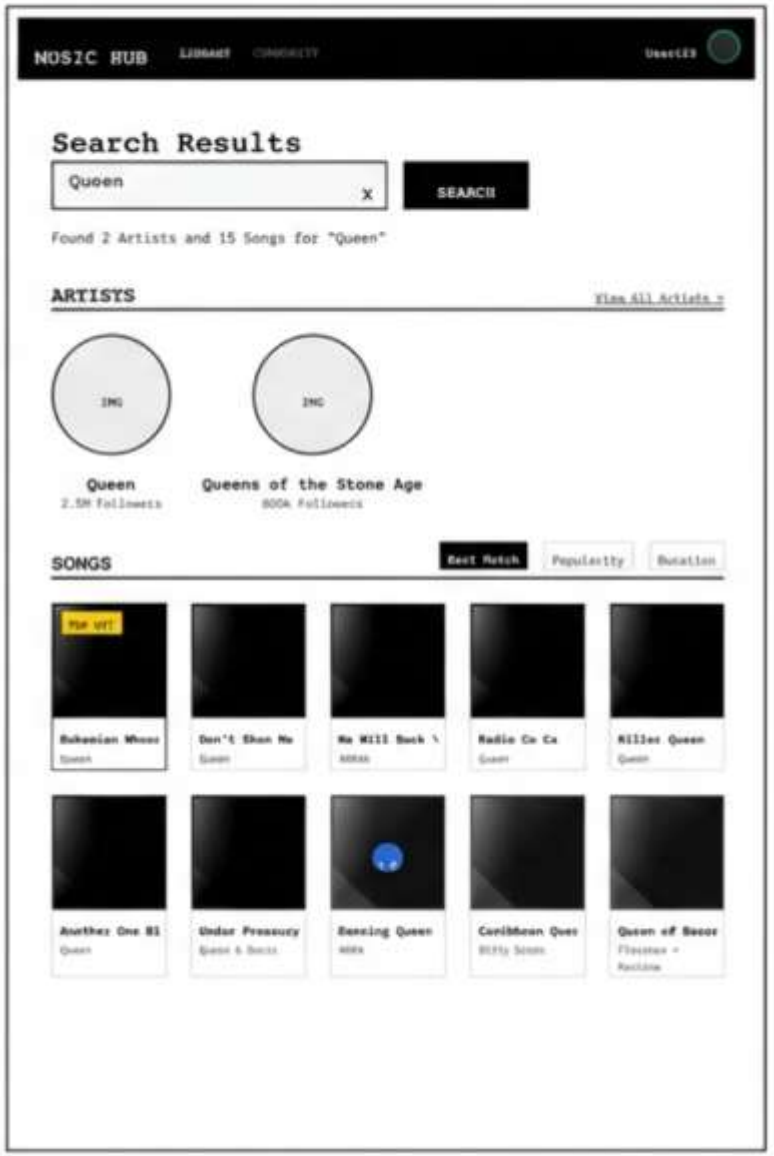
Feedback

- **id (PK):** Unique identifier for each feedback entry

Wire Frame Design of Authorisation Interface

Discovery & Navigation





Playback & Content Interaction

MUSIC HUB LIBRARY COMMUNITY User123 (U)

Library > Rock > Bohemian Rhapsody

Bohemian Rhapsody#01

Queen • A Night at the Opera (1975)

PERFORMER:	Freddie Mercury (Queen)	GENRE:	Progressive Rock
LYRICIST:	Freddie Mercury	LANGUAGE:	English
COMPOSER:	Freddie Mercury	LABEL:	EMI Records

▶ PLAY NOW + ADD TO PLAYLIST ♥ 🔗

COMMENTS (1,204)

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POST COMMENT

IMG **MusicLover99** 2 hours ago
This song never gets old. The transition between the opera section and the hard rock part is legendary!
Reply Like (45)

IMG **RockFan_Official** 1 day ago
Does anyone have the full lyrics for the middle part? Looking for the Italian phrases.
Reply Like (12)

DETAIL SPEC NOTES

Album Art
High-res Image (560x2000px min). Click expands to min). Click expands to view full size.

Metadata Gold
Dynamic fields pulled from DB:
- Performer (演唱者)
- Genre (風格)
- Lyricist (作詞)
- Composer (作曲)
- Language (語言)
- Record label (唱片公司)

Primary Actions
Buttons must be prominent. "Add to Playlist" triggers a modal to select user's existing playlists or create new.

Comment Input
Constraint: Only visible to logged in users. Guests see a "Login to comment" button here.

Comment Thread
Ordered by Date (Desc) or Popularity (Likes). Supports nested replies (1 level deep).

↓ NOW PLAYING ...

Bohemian Rhapsody

Queen

A Night at the Opera

♥ ↓ +

Mama, just killed a man

Put a gun against his head

Pulled my trigger, now he's dead

Mama, life had just begun

But now I've gone and thrown it all away

Mama, ooh, didn't mean to make you cry

01:45 05:55

⏮ ⏸ ⏭ SQ

User Personalization

User Profile

AVATAR

DISPLAY NAME

User123

LOCATION

New York, USA

BIO

Wite your text here...

EMAIL (READ ONLY)

user@example.com

DATE OF BIRTH

1995-08-24

SAVE CHANGES | CANCEL

40px

My Playlists

MY PLAYLISTS

- Liked Songs
- 2022 Top Hits
- Workout Energy
- Chill Vibes
- Road Trip

CREATE PLAYLIST

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History

BROWSING HISTORY

CLEAR ALL HISTORY

TODAY
3 SONGS

- Bohemian Rhapsody — Queen — Played 20 mins ago
- Don't Stop Me Now — Queen — Played 1 hour ago
- As If It's Your Last — BLACKPINK — Played 3 hours ago

YESTERDAY
2 SONGS

- Levitating — Dua Lipa — Dec 20, 14:30
- Bad Habits — Ed Sheeran — Dec 20, 09:15

Administration Portal

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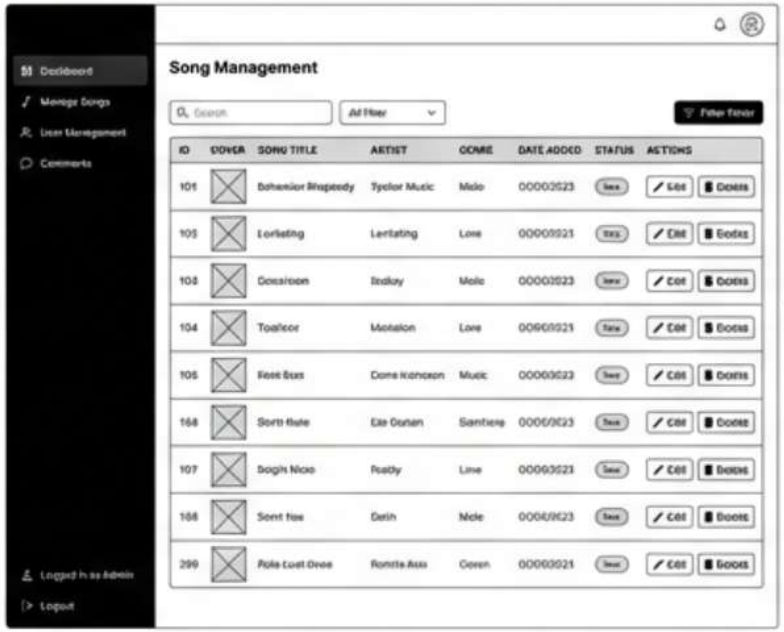

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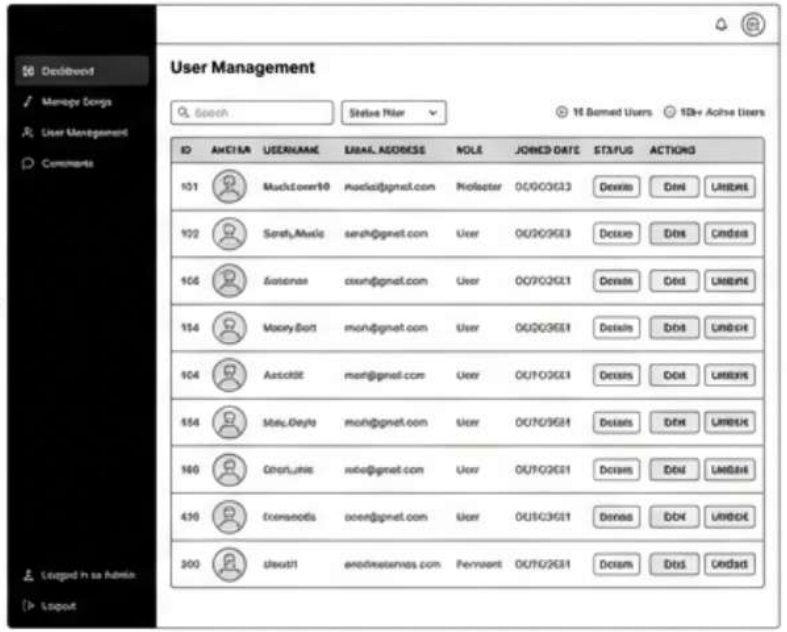
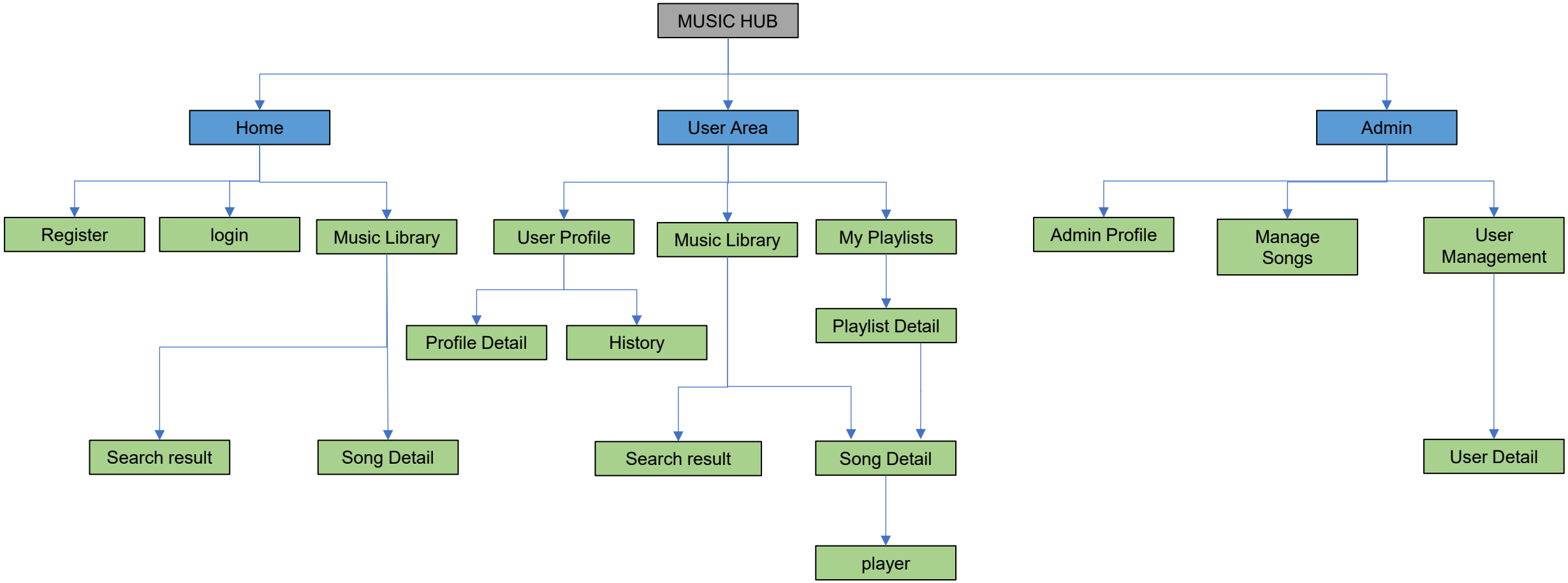


Site Map


MUSIC HUB
Home
User Area
Admin
Register
login
Music Library
User Profile
Music Library
My Playlists
Admin Profile
Manage Songs
User Management
Profile Detail
History
Playlist Detail
Search result
Song Detail
Search result
Song Detail
User Detail
player

Accessibility Plan

Category	WCAG 2.2 Success Criterion	Implementation Strategy
Colour & Contrast	1.4.3 Contrast (Minimum)	Ensure all body text has a contrast ratio of at least **4.5:1** against the background for readability.
Keyboard & Focus	2.4.7 Focus Visible	Implement a **highly visible focus ring** (outline) on all interactive elements (buttons, links) for keyboard users.
Structure & Navigation	2.5.8 Target Size (Minimum)	Ensure all clickable targets are at least **24x24 CSS pixels** to prevent accidental clicks.
Forms & Errors	3.3.2 Labels or Instructions	Provide clear, persistent **visual labels** for every input field to guide users when filling out forms.
Feedback & Status	4.1.3 Status Messages	Ensure success or error notifications are announced by **screen readers** without moving the user's focus.

Appendix

All team members actively participated in the brainstorming, design, and documentation phases of this project.

Team Member Name	Contribution (%)	Primary Responsibilities
Junyi Zhang	33%	UI Design (Wireframes), User Stories, Final Review, Documentation & Formatting
Yiyu Mao	33%	Database Design (ER Diagram), Accessibility Planning
Zijian Feng	33%	Frontend Logic, Functional Requirements, Sitemap

"We acknowledge the use of AI tools solely to enhance the visual presentation of our diagrams and refine the UI styling. All core concepts, functional requirements, and architectural decisions remain the original work of the team BE."